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AN ANALYSIS OF THE ADVANTAGES OF A BOX ISOLATION TECHNIQUE

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Résumé - Dans ce papier nous présentons un procédé BOX et les resultats électriques obtenus avec cette technique. Il utilise un contre-masque de polysilicium plutôt que d'utiliser un procédé avec une double couche de résine. Les caracteristiques électriques des transistors parasites, le courrant de fuite des diodes avec périmètres variables et la variation de la largeur de canal sont presentés.

Abstract - In this paper we present a BOX process and the electrical results obtained with this technique, in which a polysilicon counter-mask is used instead of a double resist layer to perform the field oxide. The electrical characteristics of active and parasitic transistors, the leakage current measured on diodes with various perimeters and the channel width measurements are presented.

1 - INTRODUCTION

In recent years, new isolations technologies have been developed to overcome the limitations presented by the conventional LOCOS in VLSI circuits. Most of them are based on the use of a thermal field oxide. However BOX technologies([1]) use a deposited oxide and therefore allow to get rid of the problems related to bird's beak formation, channel-stop implantation encroachment and stress generation. In order to eliminate the oxide deposited on the active areas, an additional photolithographic step seems suitable /1,2/. In this work we continue to develop a polysilicon counter-mask process/2/. Even if the BOX process is more complicated than a LOCOS or SILO one, it has some advantages that are discussed in this paper.

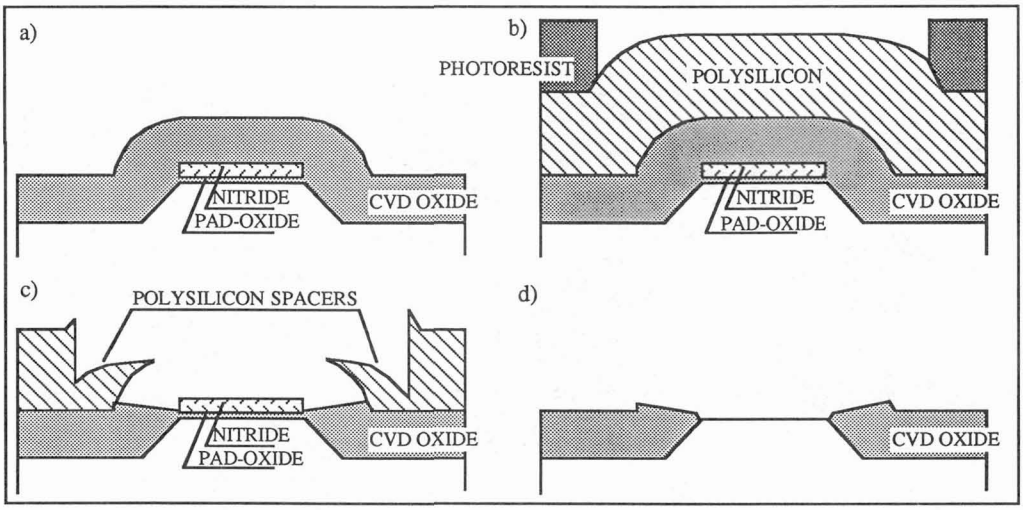


Fig. 1 - Fabrication steps of the BOX process using a polysilicon counter-mask .

([1]) The autors use the therm BOX to nominate all processes which use a deposited oxide as field oxide.

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2 - PROCESS DESCRIPTION

The starting material is <100> orientated silicon, 18Ω.cm p-wafers. First, a pad-oxide is grown and a nitride layer is deposited. After a photolithographic step to delineate the active regions, both layers are etched. Then grooves are etched in the silicon substrate with a KOH solution up to a depth of 0.5 µm, using the nitride film as a mask. After the channel-stop implantation, a SiO2 layer, with an equal thickness (or little more) to that of the Si groove depth, is deposited by CVD (Fig.1.a). A thick polysilicon layer is then deposited and whith an additional photolithographic step, a photoresist mask is left in the field areas (Fig.1.b). Then the exposed polysilicon is etched by RIE up to the oxide surface. The anisotropic etching allows the formation of spacers . Fig.1.c shows the structure after the wet etching of the oxide. During the last step, the nitride, the polysilicon and the pad-oxide layers are removed to obtain the desirable structure, as shown in Fig.1.d.
Fig.2 shows a SEM cross-section of an intermediate step of the fabrication process. In this picture we can see that even if a certain misalignement (less then 0.7 µm) of the resist mask exists, the polysilicon counter-mask is well formed. Fig 3 shows a SEM cross-section of a BOX structure with W= 1.3µm. No bird's beak is created and therefore no reduction of the dimensions are observed.

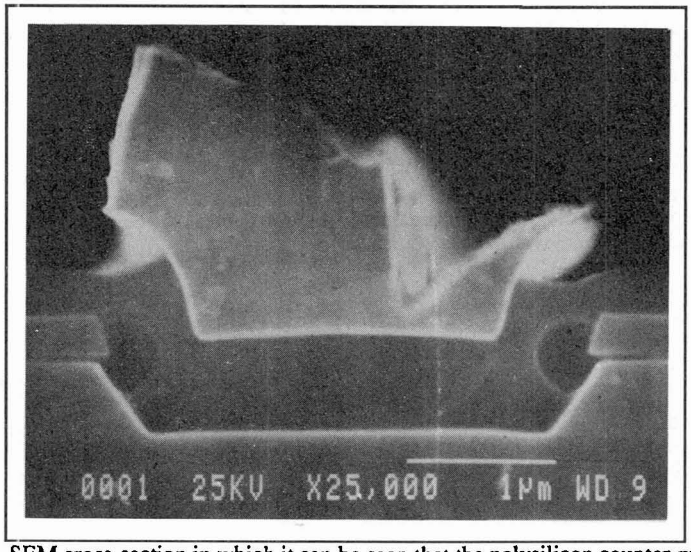


Fig. 2 - SEM cross-section in which it can be seen that the polysilicon counter-mask is auto-aligned in a range of 0.7µm

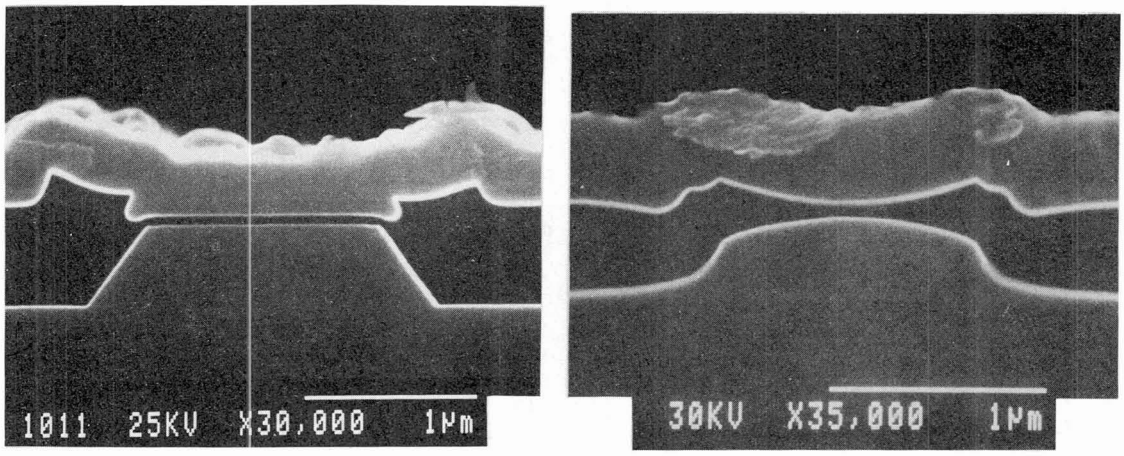


Fig. 3 - Cross-section of a 1.3µm device. Left : BOX, right : LOCOS

3 - RESULTS AND DISCUSSION

Due to the absence of thermal oxidation, the devices fabricated with a BOX process have noreduction of the active area dimensions. That is especially evident when we compare a BOX structure with a LOCOS one, as in Fig. 3. Both structures were made with the same mask. In this case the bird's beak formation in the LOCOS structure has rendered useless the active area. If we compare the BOX process with a SILO one /3/ the dimension loss is comparable, but, due to the high temperature step necessary to grow the SILO field oxide, the boron encroachment is more important with the SILO process, and so the electrical channel width is reduced. Experimentally, the effective channel width is determined by plotting the drain current of the MOS transistors (at Vd = 50mV) v.s. design mask width. The intercept with the abscissa yields a correction ΔW shown in Table I (The SILO values are obtained from ref.4)

	Φ at/cm2	Vt (V) field ox.	ΔW(1) (μm)	ΔW(4) (μm)	Δ(ΔW) (μm)
BOX	2E12	12.4	0.33	0.25	0.08
SILO	2E13	12.5	1.1	0.6	0.5

Table I. - ΔW(1) is the channel width variation for Vg-Vt = 1V
ΔW(4) is the channel width variation for Vg-Vt = 4V
Δ(ΔW) = ΔW(1) - ΔW(4)

These values demonstrate that the BOX process provides a smaller loss of channel width due to a minor boron encroachment.
The subthreshold characteristics shown in Fig.4 correspond to polysilicon gated parasitic field transistors with a width of 100μm and a length of 2 and 10μm . As it can be seen, both 2 and 10μm length field transistors have the same subtreshold characteristic. One explanation of this behaviour can be that, as demonstrated by S.H. Goodwin et al./5/ , the smaller is the radius of the transition region , the better is the subthreshold characteristiccontrol by this region. Therefore it is independent of the length of the field region.
The leakage currents have been mesured on N+P diodes with the same area (A = 1.6E-3cm2) and three different perimeters (P1 = 0.16cm, P2 = 0.96cm, P3 = 4.8cm). These currents are plotted in Fig.5. As expected, the leakage current is very low.

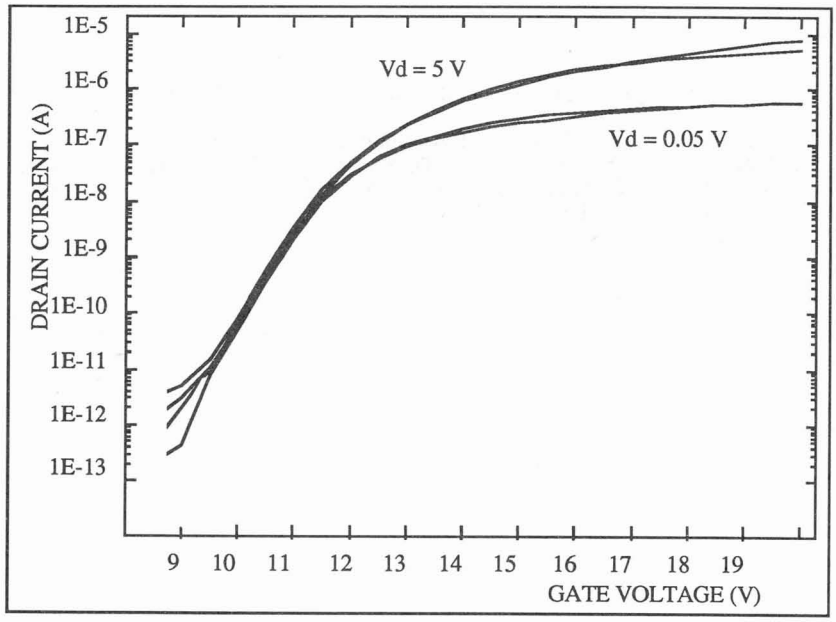


Fig.4 - Subthreshold characteristics of parasitic field transistors with L = 2μm and L= 10μm

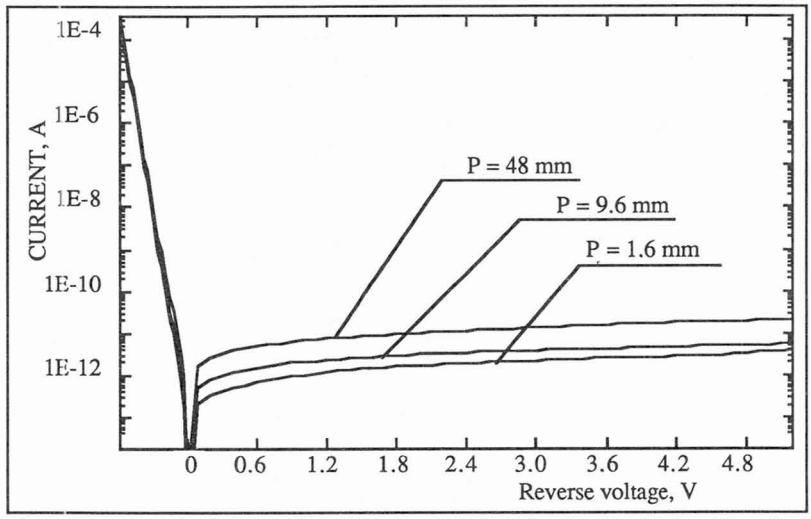


Fig. 5 - Leakage current of N^+P diodes with the same area and diferent perimeters.

CONCLUSIONS

In this work we have presented some electrical results obtained from devices fabricated with a BOX process based on the use of a polysilicon counter-mask. Even if the BOX process is more complicated than a LOCOS or SILO one, it presents some advantages : bird's beak free structures, boron encroachment reduction, very good isolations properties and a low leakage current; these adventages are related to the elimination of the long high temperature step necessary to grow a thermal field oxide.

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